May - 9 React Native Init: Core Components

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Type of mobile apps

native
↓
can work on either android or ios

cross-platform
↓
works both on android and ios

PWA
↓
Progressive Web Apps (not exactly an app)

Coding languages used:

1. Native Apps: Kotlin, Swift
2. Cross-Platform Apps: Flutter (Dart)
   React Native (JS)

npx create-expo-app@latest --template default@sdk-55

npx expo start

Download scrcpy.org

⇒ Do "~~npm~~ npm run reset-project" to remove unnecessary files.

⇒ Inside 'index.tsx'

```
import {View, Text} from "react-native";
export default funⁿ HomeScreen(){
return (
```

<View>
<Text> Hello Jaani </Text>
</View>

Image Component (self closing)

Ways to render image

1. To display image from a remote server / Image from internet

<Image
source = {{ url : "https://...
}}
width = {200}
height = {200}
/>

} As size is unknown you must and should specify size

2. For local image

<Image
source = { require ~~("@latest~~ ("@/assets/Images/..."}}
style = {{
height : 100,
width : 100,
}}
blurRadius = {30}

Text Input component


```
import { Text Input } from "react-native";

<TextInput placeholder = "enter your name" value={name}
  onChangeText - {setName}
  placeholderTextColor = {"blue"}
  />
```

Pressable Component

```
import {Pressable} from "react-native";
<Pressable onPress = {() => alert ("Button Pressed")}>
<Text > PressMe </Text>
</Pressable>
```

Adding style on pressable component, different events are also available

```
<Pressable
  onPress = {() => alert ("Button pressed")}
  style = {({ pressed }) => ({
    background Color : pressed ? "#4a42d4" : "6cF63ff"
  })}
  hit Slop = {{
     top: 10,
   bottom: 20,
  }}
  {({ pressed }) =>
     pressed ? <Text> Pressing </Text> : <Text> Press Me
                                          </Text>
  }
```

hitSlop (Increases the touchable area of a button)

To create default component (rnfe)

Inside 'index.tsx'

```
import React from "react";
import {Stylesheet, Text, View, ScrollView} from "react-native"

const HomeScreen = () => {
  const items = Array.from({length: 20},
                 (_, i) => `Item ${i+1}`);

return (
    <View>
      <ScrollView>
        {items.map((item) => (
          <Text key = {item}> {item} </Text>
        ))}
      </ScrollView>
    );
};
export default HomeScreen;
```

Switch component

```
<Switch value = {isDarkMode} onValueChange =
                                       {setIsDarkMode}
trackColor = {{false: "#d0d0d0", true: "#663ff"}}
thumbColor = {"yellow"}
/>
```

Inside "index.tsx"

```
import React, { useState} from "react";
import {
  Button,
  ScrollView,
  StyleSheet,
  Switch,
  Text,
  View,
} from "react-native";

const Home Screen = () => {
  const items = Array.from ({ length :5 }, (_,i) =>
                    `Item ${i+1}`);

  const [isDarkMode, setIsDarkMode] = useState (false);

  console.log ("hello");
  return (
    <ScrollView
      style={{ flex:1 }}
      contentContainerStyle={{
      padding :16,
      alignItems: "center",
    }}
    >
    {items.map(( item) => {
    <View
      key={item}
      style ={{
```

```
        backgroundColor : "white",
    }}
  >
  <Text style ={{ fontSize : 16 }} > {item}</Text>
  </View>
  ))}
  {Button
     title = "Hello I am a button"
     color = {"green"}
     onPress = {() => alert ("Hello Jaani !")}
  />
  <Switch
    value = {isDarkMode}
    onValueChange = {setIsDarkMode}
  trackColor = {{ false : "#ddd", true : "#6c63ff"}}
  thumbColor = {"yellow"}
  />
  </ScrollView>
 );
};
export default HomeScreen;
const styles = StyleSheet.create({});
```

Insted of <ScrollView> use <FlatList> for large list.

As ScrollView renders everything at once
<FlatList> renders items currently visible on screen.

Always keep render component seperately

<KeyboardAvoidingView> prevents the keyboard from covering the input Text.

This has a "behaviour" prop. Import "platform",

<KeyboardAvoidingView style={{ flex : 1 }} behavior= . . . .
. . . . . . }

Two common props of <KeyboardAvoidingView> are

For IOS → "padding"

Android → "height"

React-Native Safe Area Context

This is a library used in React-Native, so that UI does not overlap with,

eg:

```
<SafeAreaView style={{ flex:1 }} >
  <KeyBoard AvoidingView> . . . . . . . . . . . .
  </KeyBoard AvoidingView>
</SafeAreaView>
```

Assignment : Create a small login page inspired by the dribble ui shared.

Follow:

X: jaaani404

Happy Learning!